



AstraX Critical Path & EB1 Demonstration Readiness

A structured engineering roadmap covering full-system integration, supply chain risk control, and prototype readiness — aligned to AstraX EB1 demonstration milestones.

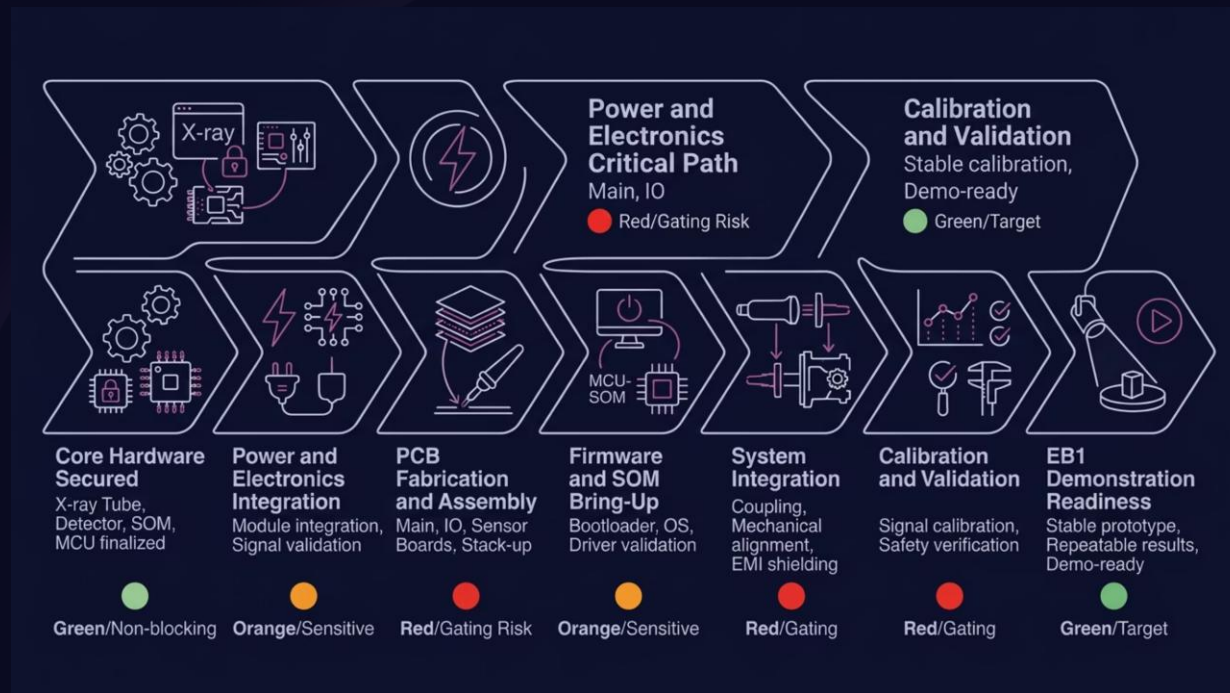


EB1 READINESS

PROGRAM REVIEW

Full System Critical Path — End-to-End

The AstraX program follows a tightly sequenced integration path from secured hardware through calibration and test infrastructure to EB1 demonstration readiness. Gating nodes must not slip.



Risk Legend

- **Gating** — Program stops if delayed
- **Sensitive** — Can delay timeline
- **Parallel** — Non-blocking execution

Key Principle

Gating nodes at PCB Fabrication, System Integration, and Calibration represent the three highest-risk program stops. All three require parallel execution tracks and early vendor engagement to protect the EB1 timeline.

Supply Chain Readiness & Component Status

Critical component acquisition is approximately **70–80% finalized**. All risk items have been identified and assigned. Power modules and PCB stack-ups remain the two highest-priority open actions requiring immediate resolution.

Component	Status	Risk	Action
X-ray Tube	✔ Locked	● Low	Proceed to integration planning
Detector	✔ Locked	● Low	Proceed to integration planning
SOM (i.MX8+)	✔ Confirmed	● Medium	Firmware alignment required
MCU (STM32)	✔ Confirmed	● Medium	Firmware development in progress
Power Modules	⚠ In Progress	● High	Source + validate immediately
PCB Stack-ups	⚠ In Progress	● High	PCB vendor finalization required

100%

Critical Components Identified

Full BOM visibility achieved across all system blocks

~75%

Components Finalized

Remaining items concentrated in power and PCB

2

High-Risk Open Items

Power Modules and PCB Stack-ups require immediate closure

Active Risk Mitigation Framework

Parallel Sourcing Strategy

Multiple qualified vendors are being engaged simultaneously for power modules and PCB fabrication to eliminate single-source dependency and reduce lead-time risk on the critical path.

Second-Source Validation

Alternate component qualification is underway for all medium- and high-risk items, ensuring program continuity in the event of supplier disruption or part obsolescence.

Early PCB Vendor Engagement

PCB fabrication vendors are being onboarded ahead of final stack-up sign-off to compress manufacturing lead times and allow design feedback loops before Gerber release.

Integration-First Test Approach

System-level integration testing is designed to begin as soon as individual subsystems clear bring-up, avoiding sequential bottlenecks and surfacing interface issues early.

- ✓ All critical supply chain and system integration risks have been identified, assigned, and are under active execution control — aligned to AstraX EB1 demonstration timelines.

Execution Discipline Driving EB1 Readiness

AstraX program leadership is operating with full component-level visibility, real-time risk tracking, and a unified supply chain and engineering execution model — the foundation for a credible, repeatable EB1 demonstration.



Component-Level Visibility

Every subsystem block — from tube and detector to power modules and PCB stack-ups — is tracked against a defined status and risk tier, ensuring no hidden gaps in the BOM or integration plan.



Critical Path Actively Tracked

The full end-to-end critical path from hardware lock through calibration and demo readiness is identified, sequenced, and owned — with gating nodes monitored against EB1 milestone dates.



Supply Chain + Engineering Unified

Supply chain acquisition and engineering integration are operating as a single execution unit — procurement decisions are made with firmware and system integration requirements directly in scope.

EB1 Target State: A stable, demo-ready AstraX prototype delivering repeatable, validated results — backed by full traceability from component sourcing through calibrated system performance.